

PHILIP UMEADI

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EDUCATION

Syracuse University

College of Arts & Sciences

Bachelor of Science in Applied Mathematics, Minor in Economics

Syracuse, NY

August 2023 – May 2026

3.16

RELEVANT WORK EXPERIENCE

Saint Joseph's Preparatory School, Math Teacher, Philadelphia, PA

August 2026 – Present

- Teaching algebra and geometry courses under tenured math faculty, and supporting students beyond the classroom

Verisk, Analyst Intern, Jersey City, NJ

June 2025 – August 2025

- Collaborated with senior developers to engineer a scalable Python framework (pandas vectorization, regex, RapidFuzz and pypdf) to complete a vendor record migration of 27,000+ legacy documents to a new database ahead of contract renegotiations.
- Built a Python automation suite that replaced hand-editing recurring vendor correspondence

Extraordinary Re, Research Intern, New York, NY

May 2023 – August 2023

- Researched how a consortium model could attract institutional participants to a marketplace for trading insurance contracts and recruited a former GE Private Equity VP to spearhead operations

RESEARCH & PROJECTS

Applied Research Tasks, NYU Stern Research Assistant Application

- Drove research and engineering tasks using AI coding agents in sandboxed Docker environments

Mathematical Foundations of Machine Learning & NLP, supervised by Justin Ko, PhD, Syracuse University

- Conducted a supervised reading course through three standard machine-learning texts (Deisenroth; James; Jurafsky & Martin), progressing from linear regression and PCA to the mathematics of neural-network training, and implementing selected methods in Python.
- Developed a nonlinear regression model in Python to forecast a year of natural gas prices using polynomial-sinusoidal basis functions and MLE. I chose the model's complexity by testing forecast error on held-out data, so the model captured the seasonal pattern rather than memorizing the past 48 months.
- Built a character-level n-gram language model in Python to play Hangman on a 370K-word corpus. By linearly interpolating unigram through trigram models with perplexity-optimized mixture weights, the model achieved 8.2 average incorrect guesses per word, beating a letter-frequency baseline of 14. I showed the model automatically leans on simpler statistics when it has less data to work with. Paper and code available here: <https://github.com/mindphil/mml/tree/main/hangman>

Binary Fission as a Homogeneous Branching Process, Stochastic Processes Independent Project

- Modeled bacterial colony extinction under antibiotic treatment as a homogeneous branching process. Derived a closed-form threshold for the minimum drug concentration guaranteeing extinction, then compared the theoretical efficacy of two real antibiotics using pharmacodynamic parameters from the literature (Bogdanov et al.). I then built a Python/Manim simulation that animates the stochastic process. Paper and code available here: <https://github.com/mindphil/branching-process-fission>

Combinatorial Game Theory & C++ Systems Programming

- Investigated a network pursuit game (an n-cop generalization of the Shannon Switching Game, due to K. Wood) and conjectured that the resources needed grow linearly with network size, far less than the quadratic bound previously established; presented at a seminar (Jan. 2026).

Linear Algebra and Category Theory, supervised by Ben Kaufman, Syracuse University

- Studied how large linear systems break into irreducible building blocks (the Kronecker problem).

TECHNICAL SKILLS

Programming: Python, C++ (basic), MATLAB, web scraping (BeautifulSoup)

Tools: Git, Docker, REST/LLM APIs (Claude, Gemini, Groq), Bloomberg Terminal, Excel/VBA

Mathematics: Probability theory, stochastic processes, numerical methods, (applied) linear algebra, ODEs & PDEs

LEADERSHIP & ACTIVITIES

Syracuse University Division 1 Rowing

August 2023 – June 2026

- 20+ hours a week of training while carrying a full applied-math course load. Finished my final season with a 4th place finish at our championship regatta and an undefeated regular season

Manhattan College FED Challenge & Investment Club

September 2022 – May 2023

- Built econometric models forecasting CPI, unemployment, and GDP
- Conducted DCF and comparable company analysis on equity positions for student-managed portfolio